Training Day 17

1. Topics Covered and Brief Explanation

1. Introduction to Deep Learning: Deep Learning is a branch of Machine Learning that uses artificial neural networks to learn patterns from large amounts of data.
2. Machine Learning vs Deep Learning: Traditional Machine Learning often requires manual selection of important features, whereas Deep Learning can automatically learn useful features from data.
3. Artificial Neural Networks (ANN): ANNs are computing models inspired by the structure of the human brain. They consist of interconnected units called neurons.
4. Layers of a Neural Network:
    - Input Layer: Receives input data.
    - Hidden Layer: Processes information and identifies patterns.
    - Output Layer: Produces the final prediction or result.
5. Neurons and Weights: Each neuron receives inputs, assigns weights to them, and calculates an output using a bias and an activation function.
6. Activation Functions: Activation functions help neural networks learn complex patterns. Common examples include ReLU and Sigmoid.
7. Applications of Deep Learning: Image recognition, speech recognition, facial recognition, language processing, and self-driving vehicles.

2. Tools and Technologies Used

Tool	Purpose
Python	Programming language
NumPy	Numerical calculations
Google Colab	Writing and executing Python code
Neural Networks	Understanding deep learning concepts

3. Practical Learning / Code Examples

A basic neuron calculation was performed using NumPy to understand how inputs, weights, and bias contribute to the output.

```
import numpy as np

inputs = np.array([0.5, 0.8])

weights = np.array([0.4, 0.6])

bias = 0.1
```

```
weighted_sum = np.dot(inputs, weights) + bias

# ReLU activation function
output = max(0, weighted_sum)

print("Neuron Output:", output)
```

Explanation:

- inputs represents the values given to the neuron.
- weights represent the importance assigned to each input.
- np.dot() calculates the weighted sum.
- bias is added to the weighted sum.
- ReLU returns the value if it is positive; otherwise, it returns zero.

4. Learning Outcomes

- Understood the basic concept of Deep Learning.
- Learned the structure and working of Artificial Neural Networks.
- Identified the functions of input, hidden, and output layers.
- Understood the role of weights, bias, and activation functions.
- Explored practical applications of Deep Learning.